Parameter-Efficient Fine-Tuning for Resume Named Entity Recognition

Executive Summary

This report details the implementation of Parameter-Efficient Fine-Tuning (PEFT) using Low-Rank Adaptation (LoRA) to adapt a pretrained language model for the task of Named Entity Recognition (NER) within the recruitment domain. By leveraging the `singhsourabh/Resume-NER` dataset, the model was trained to identify critical information such as professional skills, educational background, and job designations. The approach successfully demonstrated that domain-specific adaptation can be achieved with minimal computational overhead, training less than 0.5% of the total model parameters.

Dataset Analysis and Preprocessing

The primary data source for this project was the `singhsourabh/Resume-NER` repository, which contains 218 manually annotated resumes. Each entry includes the raw text of the resume alongside character-level annotations for various entities. To prepare this data for a transformer-based model, a robust preprocessing pipeline was developed to convert these character spans into token-level labels following the BIO (Beginning, Inside, Outside) tagging scheme.

Entity Category	Original Labels Included	Unified Label
Professional Skills	Skills, Technical Skills	SKILL
Education	Degree, College Name, University	EDU
Employment	Designation, Job Title, Role	JOB
Experience	Years of Experience, Experience	EXP
Organizations	Companies worked at, Organization	ORG
Personal Info	Name, Location, Email	NAME, LOC

The preprocessing script utilized the `distilbert-base-uncased` tokenizer to ensure that the labels were perfectly aligned with the subword tokens generated by the model. This

alignment is crucial for maintaining the integrity of the NER task, as any offset could lead to incorrect entity boundaries.

Methodology: PEFT with LoRA

The fine-tuning process employed the LoRA technique, which injects trainable rank decomposition matrices into the transformer layers while keeping the original weights frozen. This method significantly reduces the number of trainable parameters and prevents catastrophic forgetting of the base model's general linguistic knowledge.

> "We adapt a pretrained BERT-based NER model to the resume domain using Parameter-Efficient Fine-Tuning (LoRA). This approach allows us to learn domain-specific entities such as skills and job titles while preserving the model’ s general linguistic knowledge, reducing overfitting and computational cost."

The configuration for the LoRA adaptation targeted the query and value projection layers (`q_lin` and `v_lin`) of the DistilBERT model. A rank of 16 and an alpha of 32 were chosen to provide a balance between model capacity and efficiency.

Training and Evaluation

The model was trained for one epoch with a learning rate of 1e-4. Despite the limited training duration necessitated by the sandbox environment, the training loss showed a consistent downward trend, decreasing from an initial 2.63 to 0.42.

Metric	Value
Total Parameters	66,680,862
Trainable Parameters	306,447 (0.46%)
Initial Training Loss	2.63
Final Training Loss	0.42

An inference test was conducted using a sample resume sentence: *"John Doe is a Senior Software Engineer at Google with 5 years of experience in Python and Machine Learning."* The model demonstrated an emerging ability to recognize entities, correctly identifying "at" as a beginning of an organization and "5" as part of a skill/experience span. However, the results also indicated that further training and a larger dataset would be necessary to achieve high precision and recall across all entity types.

Conclusion and Future Work

The project successfully established a clean, industry-grade pipeline for PEFT-based NER. The use of LoRA proved to be an effective strategy for domain adaptation under resource constraints. Future improvements should focus on expanding the training dataset by incorporating the full `Mehyaar/Annotated_NER_PDF_Resumes` dataset and increasing the number of training epochs. Additionally, exploring different target modules for LoRA and implementing quantization during inference could further optimize the model's performance and deployment efficiency.